Campus-DDoS-SDN: A Python-based testbed for multi-vector DDoS attack detection in software-defined campus networks

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ABSTRACT

Campus-DDoS-SDN is a Docker testbed that generates and classifies four DDoS attacks against the portal of a campus emulated with Mininet and Ryu. Students keep browsing during the attack, and each flow is labelled by source IP rather than by capture file. A random forest trained without IP addresses and tested on the Ciencias Agrarias laboratory, which was held out of training, reaches a macro F1 of 0.979. On the portal probe, httpflood: the page slows down; packet rate rises; icmpflood: packet rate at the portal rises; slowloris: the page stops opening; synflood: packet rate at the portal rises. ICMP is not claimed as a settled class: the Agrarias test contains 24 flows.

1. Introduction

Software-defined networking (SDN) decouples the control plane from the data plane, concentrating network intelligence in a logically centralized controller [1]. This centralization simplifies network management, but it also turns the controller into a single point of failure: a distributed denial-of-service (DDoS) attack that saturates the controller, the switches, or a hosted service can degrade or take down the entire network segment it governs. Detecting such attacks early, and separating them reliably from legitimate traffic spikes, requires datasets that are both representative of real SDN deployments and precisely labelled.

Most publicly available intrusion-detection datasets, such as NSL-KDD [6], CICIDS2017 [4], and CICDDoS2019 [5], were not captured inside an SDN environment: they lack OpenFlow-specific signals, such as flow-table occupancy or controller load, and are nonetheless often reused out of context to train SDN-targeted classifiers. Recent work has started to close this gap by generating DDoS datasets directly inside Mininet/Ryu testbeds. Kumar and Gupta released two single-vector datasets built this way: a TCP-SYN flood dataset [2] and, more recently, an ICMP-flood dataset [3] in which normal traffic is captured by connecting a Mininet host to a live internet connection and browsing real websites (YouTube, Facebook, news, etc.), instead of relying on synthetic traffic generators. This real-traffic approach increases realism, but it introduces two practical limitations for a reusable benchmark: the resulting dataset is tied to a single attack vector, and it is not reproducible across runs, since the captured content and timing depend on whatever the live internet happens to serve at capture time.

This work, Campus-DDoS-SDN, addresses both limitations within a single, fully reproducible testbed. Four attack vectors (TCP-SYN flood, HTTP flood, Slowloris, and ICMP flood) are launched from already-compromised hosts against one shared victim, a web portal, inside a Mininet/Ryu topology that mimics a university campus: a core/DMZ switch layout plus one emulated laboratory per academic area. Unlike prior single-vector or fully-separated-capture datasets, legitimate users keep browsing the portal throughout every attack, so each capture mixes benign and malicious flows against the same target, as a real DDoS incident would. Ground-truth labels are assigned per source IP rather than per capture file, and every run additionally logs the attack's effect on the victim as perceived by a real client (HTTP response code, latency, active connections), alongside network-level indicators (packets and bytes per second, OpenFlow flow-table size). The complete environment, including the traffic generators, the custom web portal, and the labelling pipeline, is packaged as a Docker image to remove any dependency on a live internet connection or on a specific host operating system.

The main contributions of this work are:

i. A multi-class DDoS dataset covering four attack vectors against a single victim inside a realistic, multi-segment SDN topology, instead of a single attack type against a flat network.

ii. Per-flow, per-source-IP labelling that keeps legitimate and malicious traffic mixed within the same capture, closer to a real attack scenario than the fully-separated captures used in prior work.

iii. Combined network-level (flow and packet statistics via CICFlowMeter) and application-level (HTTP status, latency, connection saturation) impact metrics, captured automatically on every run.

iv. A one-command, fully reproducible Docker environment that does not depend on a live internet connection, addressing the reproducibility limitation of real-traffic-based SDN datasets.

v. A random-forest classifier trained without IP addresses that reaches a macro F1 of 0.979on the held-out Ciencias Agrarias laboratory, plus a pipeline

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Table 1
Code metadata.

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0
C2	Permanent link to code/repository used for this code version	Fill in the URL at publication
C3	Permanent link to Reproducible Capsule	–
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python, Bash, Mininet, Open vSwitch, Ryu, CICFlowMeter, Docker
C7	Compilation requirements, operating environments & dependencies	Docker, Ubuntu 24.04; Python 3.9.25 and 3.12.3
C8	Link to developer documentation/manual	Ver `README.md` del repositorio
C9	Support email for questions	cristhianmartinezcasas@gmail.com

ready to mix authorized Wireshark captures of benign student traffic with simulation-only attacks.

2. Software description

2.1. Architecture overview

The topology is a reduced campus: an edge modem, a core switch, a DMZ holding the portal, and one laboratory per academic area. Each laboratory has students and compromised hosts. Switches are Open vSwitch in userspace, controlled by Ryu. Students keep requesting portal pages during an attack. The operator's browser, at `http://127.0.0.1:8080/`, joins as one more benign client (`10.9.0.1`). The attack does not leave the container.

2.2. Software functionalities

`scripts/docker/run_scenario.sh` runs one scenario (`baseline`, `synflood`, `httpflood`, `slowloris`, `icmpflood`) or all five. `scripts/mininet/capture_campus.py` stores the pcap and the student probe. `scripts/mininet/pcap_to_csv.py` labels each flow by source IP: a student stays `BENIGN` and a compromised host takes the attack name. `notebooks/analisis_completo.py` trains the forest without IP addresses and tests it on Ciencias Agrarias. `scripts/docker/preparar_hibrido.sh` anonymizes authorized student Wireshark captures, mixes them with simulation attacks, and writes `notebooks/resultados/informe_hibrido.txt`. The attack is never run against the real campus network.

3. Impact overview

The single-vector reference dataset cannot be repeated, because its benign traffic comes from the live internet [3]. This testbed can be repeated, mixes student and attack traffic in the same capture, and already yields a classifier. On Ciencias Agrarias, a laboratory excluded from training, macro F1 is 0.979. The probe states what the page feels, not only which label was assigned. With institutional authorization, the same pipeline admits real benign traffic (Wireshark) while keeping the attack inside the container.

Table 2
Detection (random forest). Macro F1 0.979.

Clase	Prec.	Recall	F1	Flujos	On the flow
BENIGN	0.956	0.949	0.953	1775	well separated
httpflood	0.939	0.947	0.943	1458	well separated
icmpflood	1.000	1.000	1.000	24	too few flows to claim
slowloris	1.000	1.000	1.000	126	well separated
synflood	1.000	1.000	1.000	4708	well separated

3.1. Ongoing research projects using the software

The testbed is used in the authors' undergraduate thesis at the Universidad Nacional del Centro del Perú. The Wireshark capture protocol for the network officer is documented in `docs/documento_encargado.docx`.

3.2. Limitations and future improvements

The table is still computed from emulated students. The hybrid dataset (Wireshark plus simulated attack) is ready in the code and is updated when the authorized pcaps arrive. Every scenario in this run lasts 120 s. ICMP accounts for 120 flows in the whole set and 24 in the Agrarias test, so its perfect F1 is not reported as a settled class result. The classifier does not run inside Ryu; it scores flows that have already been captured. Future work: repeat the test with the Wireshark background, a mixed four-vector scenario, and more ICMP samples.

3.3. List of scholarly publications enabled by this software

There is no other publication yet. This software is what makes the evaluation in this article possible.

4. Software and hardware requirements

The testbed runs in an Ubuntu 24.04 container. The versions in the table were read inside the `ataque-sdn` image. The model was trained with Python 3.12.3, scikit-learn 1.9.1 and pandas 3.0.6. Ryu uses Python 3.9.25.

Table 3
Testbed setup requirements.

N.°	Description	Tool	Version
1	Network emulator	Mininet	2.3.0
2	Virtual switch	Open vSwitch	3.3.9
3	SDN controller	Ryu	4.34
4	SYN and ICMP flood	hping3	3.0.0-alpha-2
5	Low-rate attack	slowhttptest	1.9.0
6	Packet capture	tcpdump	4.99.4
7	pcap to CSV	CICFlowMeter	0.5.0
8	Container OS	Ubuntu	24.04

The run reported here used an Apple M5 Pro host (15 cores, 24 GB of RAM, macOS 27.0.1) with Docker Desktop. The container was given 6 CPUs and 7.7 GiB of RAM. Open vSwitch runs in userspace (`datapath netdev`).

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